

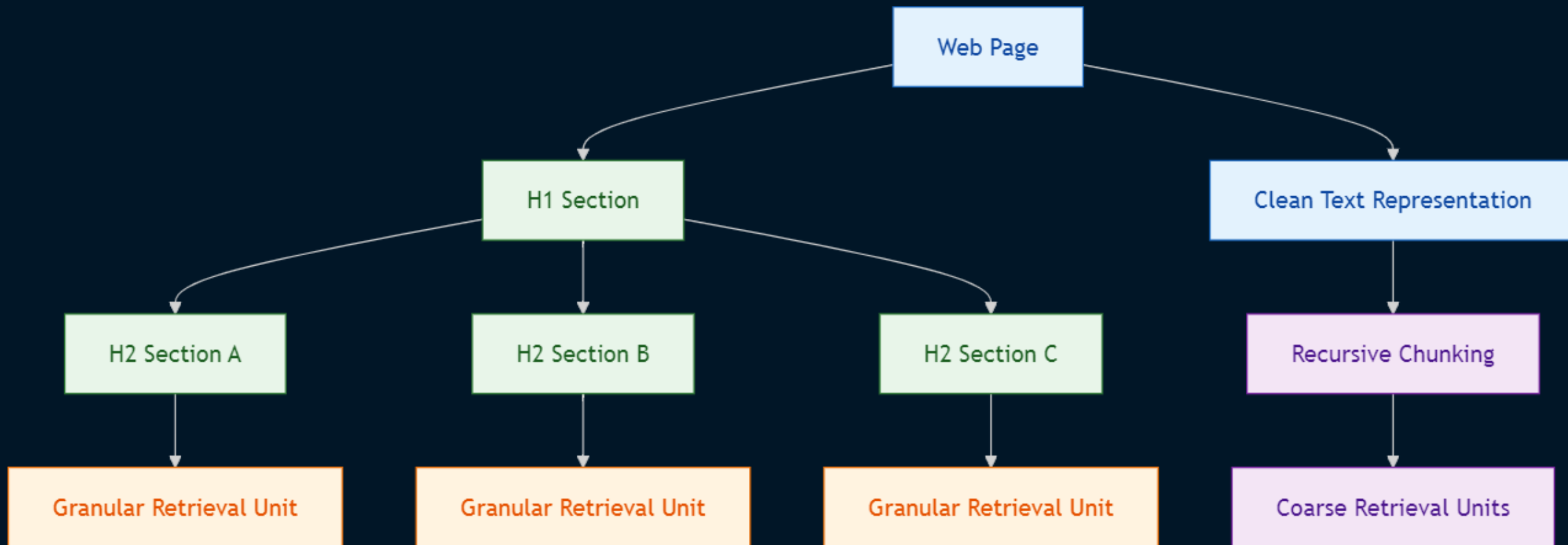
Agentic AI Engineering

***Advanced Hierarchical RAG Indexing
and
Retrieval with ChromaDB***

Max Montaseri

- Designed an **advanced hierarchical RAG indexing architecture** using HTML-aware section splitting and recursive character chunking to compare granular versus coarse retrieval strategies.
- Implemented dual ChromaDB indexes for granular and coarse document representations using OpenAI embeddings.
- Built an HTML ingestion pipeline with AsyncHtmlLoader, HTMLSectionSplitter, and Html2TextTransformer for structure-aware knowledge extraction.
- Configured RecursiveCharacterTextSplitter with 3,000-character chunks and 300-character overlap for coarse-grained semantic retrieval.
- Developed reusable ingestion utilities for transforming web documents into vector-searchable chunks and managing destination-specific Chroma collections.
- Extended the indexing experiment from a single Cornwall knowledge source to a multi-destination UK tourism corpus.
- Implemented comparative similarity-search experiments across granular and coarse indexes to investigate retrieval behavior for destination-specific questions.
- Established an extensible foundation for hierarchical retrieval, metadata filtering, reranking, citation-aware generation, and quantitative RAG evaluation.

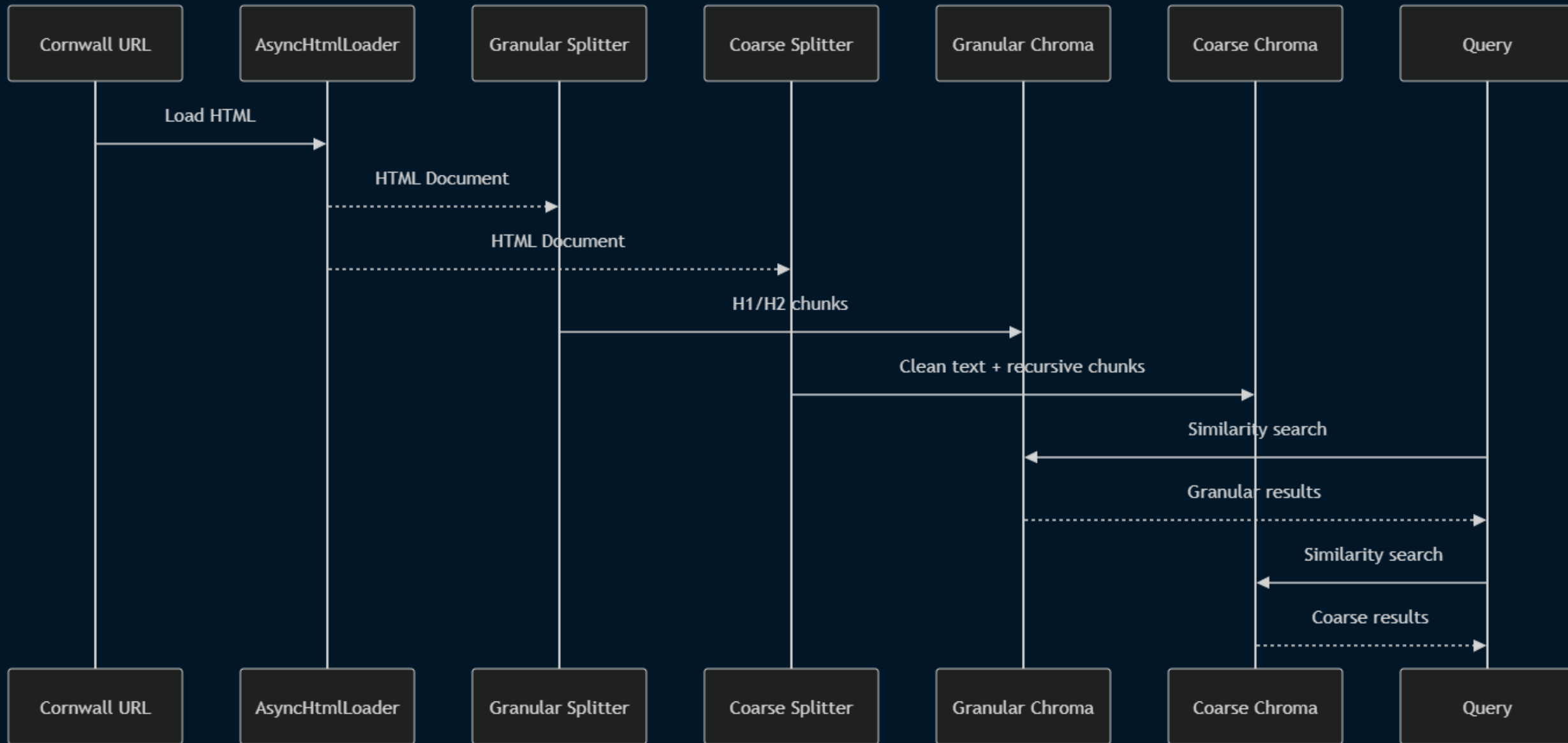
Core Concept: Document Hierarchy



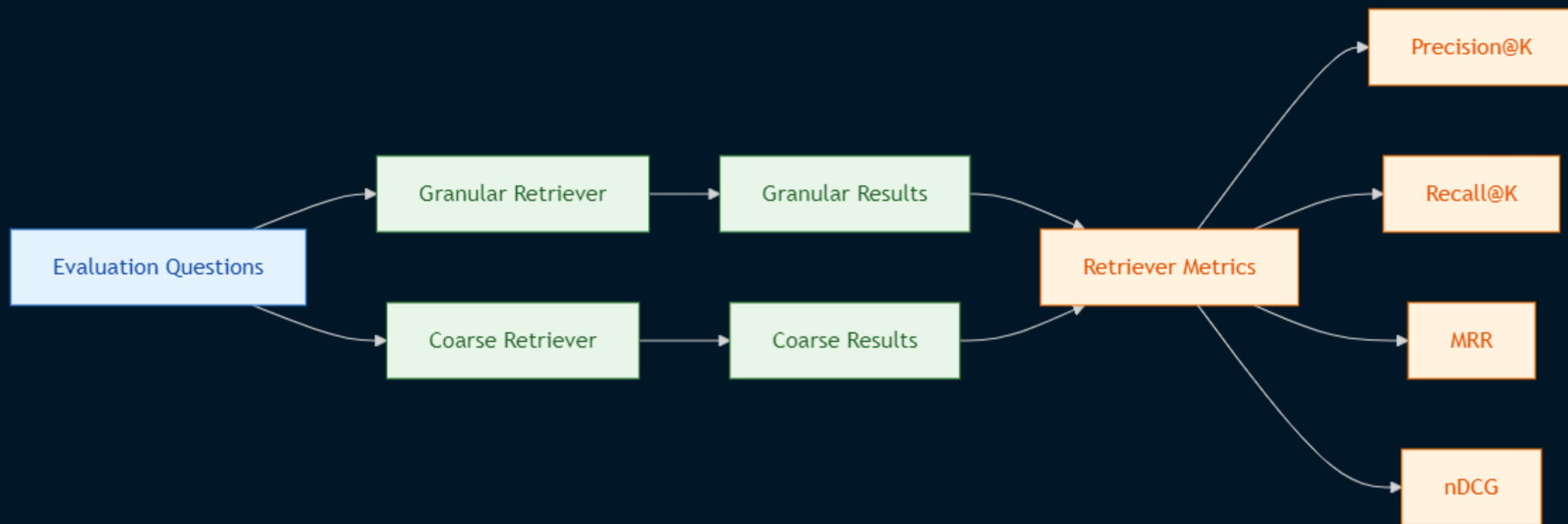
Architecture



Destination Workflow



Recommended Evaluation Architecture



Enterprise RAG Evolution

